

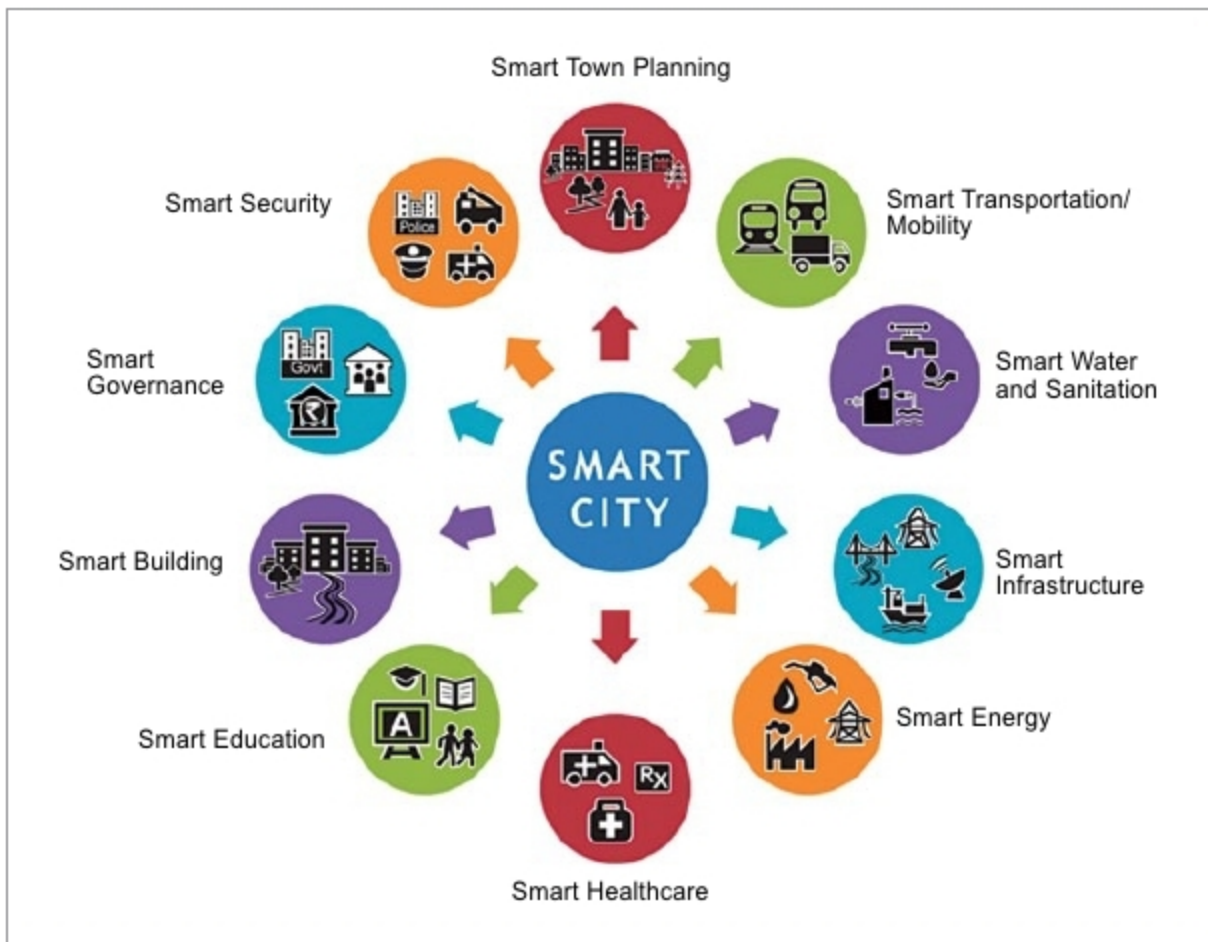


Fig. 2: Smart city development areas (Credit: www.rolta.com)

deploy, operate and monetise carrier-grade IoT connectivity on their own. These networks turn out to be key assets that cities want to control, financially and/or operationally, and that can be combined to power a growing diversity of use cases. Considering their budget challenges, cities are also looking for a better trade-off between financial sources and possible business models over time. Opting for a relevant alternative like low-power wide-area networks (LPWAN) and selecting a partner that can quickly and easily adapt to evolving collaboration models enable cities to address both their technical needs and financial challenges.”

Developing a sustainable city

To build future-ready smart cities, there are a few fundamental things that are required to establish a stable and adaptable framework.

Smart infrastructure. This is required to establish a foundation for digital economic development. Heating, energy usage, lighting and ventilation in buildings are managed and optimised by technology. Water-leakage detection and water potability monitoring are integrated into smart buildings. Rooftop gardens

or side vegetations are integrated into building designs to produce oxygen and absorb carbon-dioxide.

Smart infrastructure also includes city-wide Wi-Fi for public use to provide real-time information on traffic congestion, parking spaces and other amenities. The aim of smart parking is to reduce both car emissions and traffic.

Smart roads are prepared for autonomous vehicles and self-driving car traffic system integration. With smart drainage systems and filtration, cities could drastically improve local living conditions. Smart lighting control systems reduce energy consumption, carbon emissions and maintenance costs, while providing a safe environment.

Security. Creating cities that are affordable and safe should be the priorities. This ensures secure and efficient operation of critical applications developed for the smart cities, which rely on constant connectivity to volumes of data from stationary and moving sensors, which is transformed into useful information through data analytics to provide a better quality of life. Failure to secure this data can cause extreme damage.

Be it a residential complex, market, school, hospital, district

centre or public transport, sustenance of any urban setup is impossible without robust security measures. But with the advent of AI, the concept of a smart city is viable as well as feasible. AI has empowered security to automatically learn and detect crimes.

Sustainable energy. Solar panels are integrated into building design, replacing traditional materials. Solar and wind energy can be collected throughout the day by fully integrating these into roads, buildings and residential areas. Smart grids are used for energy consumption monitoring and management. These conserve energy in places that might not need power and then send it to areas that may need even more power.

A building can become entirely self-sufficient by using technology like solar windows. Unused energy generated can be offered to vehicles in the local area. By 2060, cargo will travel through hyperloop and will be moved rapidly around the world in smart containers that know their contents and their destination. Ports themselves will be automated, running on renewable energy and having zero carbon emissions.

Development of smart cities will require more energy, so the focus should be on sustainable energy sources. For example, cities can pave the roads with some kind of material that converts solar energy into electricity. Automobiles can be equipped with solar panels on roofs. Autonomous, environment-friendly, efficient transport can save time and money, too.

Digitalisation and automation. Smart cities use IoT devices and sensors to gather and analyse information across the infrastructure. This helps city authorities to intelligently manage their assets, increase efficiency, revolutionise transport, reduce costs and enhance the overall quality of life for residents.

Management. This is the last but the most important stage of building a smart city. It requires a platform that features AI and machine learning to monitor and improve smart city projects.